A7 Design 1

	Coarse mesh	2nd test Mid mesh	3rd test Fine mesh
	initial design		
max Stress	226.7 Mpa	304.0 Mpa	422.1 MPa
+ Location	Small corner for rod	Small corner for	Small corner
max deflect	.00424 in	.004303	.004355
+Location	free end of rod	free end of rod	free end of rod
mass	.30 lbs	.30 lbs	30 lbs
SF	3.65	2.72	1.96
# of elements	1341	8955	65177

% Stress error $= \left|\frac{304 - 226.7}{226.7}\right| \times 100 = 34.1\%$ X

% Stress = 38.85% X

% def = .73% ✓

% def error $= \left|\frac{.004303 - .00424}{.00424}\right| \times 100 = 1.39\%$ ✓

After 3 tests Stress diverging def converging

A 7 Design 2

	Coarse mesh	Med Mesh	fine mesh
Max σ & Location	407.5 Mpa at rod to Sup	405.1	connection not needed convergence hit!
Max δ & Location	.004169 in at end of rod	.004189	
mass	.30 lbs	.30 lbs	
SF	2.03	2.04	
# of Elements	1818	7910	

% Stress error = $\left|\frac{405.1-407.5}{407.5}\right| \times 100 = .59\%$ ✓

% def error = $\left|\frac{.004189-.004169}{.004169}\right| \times 100 = .47\%$ ✓ Convergence

A7 Design 3

	Coarse mesh	Med Mesh	Fine mesh
Max σ + Location	125.8 Mpa	144.7 Mpa at Support to rod connection	152.0 Mpa
Max δ + Location	.001211 in	.001237 in at end of rod	.001245
mass	.30 lbs	.30 lbs	.30 lbs
SF	6.58	5.72	5.44
# of Elements	1524	8198	63997

Stress % error $= \left|\frac{144.7 - 125.8}{125.8}\right| \times 100 = 15.02\%$ ✗

def % error $= \left|\frac{.001237 - .001211}{.001211}\right| \times 100 = 2.18\%$ ✓

Stress % error $= \left|\frac{152 - 144.7}{144.7}\right| \times 100 = 5.04\%$ ✓

displacement % error $= \left|\frac{.001245 - .001237}{.001237}\right| \times 100 = .60\%$ ✓

Convergence

	Design 1	Design 2	Design 3
Max σ + Location	111.1 Mpa at connection of rod and	105.1 Mpa	152 Mpa Support to bracket
Max δ + Location	.001256 in at the end of the	.001184 in	.001245 in rod
mass	.30 lbs	.30 lbs	.30 lbs
SF	1.96	2.04	5.44
# of elements	65177	7910	63997

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